

Application Build and Release Automation Lab

Assignment 2 — Solutions

Solutions arranged according to Question 1 → Question 5

Student: Divyansh Negi

Roll Number: 2415001880

QUESTION 1 — Linux Environment and File Management

```
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~
$ # QUESTION 1- Linux Project Setup

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~
$ mkdir Ayush-GLA-HealthSystem

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~
$ cd Ayush-GLA-HealthSystem

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem
$ mkdir src docs config backup

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem
$ touch health_info.txt

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem
$ nano health_info.txt

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem
$ pwd
/c/Users/DIVYANSH NEGI/Ayush-GLA-HealthSystem

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem
$ cd src

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem/src
$ cd ..

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem
$ ls
backup/  config/  docs/   health_info.txt  src/

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem
$ cat health_info.txt
Student Name: DIVYANSH NEGI
Roll Number: 2415001880
Subject Name: Application Build and Release Automation Lab
Project Name: Campus Health Appointment System
Semester: 5th

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem
```

```
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem
$ git init
Initialized empty Git repository in C:/Users/DIVYANSH NEGI/Ayush-GLA-HealthSystem/.git/

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git config --global user.name "DIVYANSH NEGI"
$ git config --global user.email "divyanshnegi20@gmail.com"

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    health_info.txt

nothing added to commit but untracked files present (use "git add" to track)

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git add .
warning: in the working copy of 'health_info.txt', LF will be replaced by CRLF the next time
Git touches it

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git commit -m "Initial commit for health appointment system"
[master (root-commit) e14286d] Initial commit for health appointment system
1 file changed, 5 insertions(+)
create mode 100644 health_info.txt

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ nano health_info.txt

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git diff
warning: in the working copy of 'health_info.txt', LF will be replaced by CRLF the next time
Git touches it
diff --git a/health_info.txt b/health_info.txt
index f92ed3f..a12cd85 100644
--- a/health_info.txt
+++ b/health_info.txt
@@ -3,3 +3,7 @@ Roll Number: 241501877
    Subject Name: Application Build and Release Automation Lab
    Project Name: Campus Health Appointment System
    Semester: 5th
+
+Project Objective: To develop a Campus Health Appointment System for managing health appointments.
+Student Program/Course: B.Tech Computer Science and Engineering
```

QUESTION 2 — Git Repository and Version Control

```
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   health_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git add health_info.txt
warning: in the working copy of 'health_info.txt', LF will be replaced by CRLF the next time Git touches it
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git commit -m "Add project objective and course details"
[master 9adf74e] Add project objective and course details
1 file changed, 4 insertions(+)
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git log --oneline
9adf74e (HEAD -> master) Add project objective and course details
a124286 Initial commit for health appointment system
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git status
On branch master
nothing to commit, working tree clean
```

```
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch appointment-feature
Switched to branch 'appointment-feature'
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch --show-current
appointment-feature
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ touch src/appointment.txt
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ nano src/appointment.txt
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git add src/appointment.txt
warning: in the working copy of 'src/appointment.txt', LF will be replaced by CRLF the next time Git touches it
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git commit -m "Add appointment module"
[appointment-feature 5d7e071] Add appointment module
1 file changed, 11 insertions(+)
create mode 100644 src/appointment.txt
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch
* appointment-feature
  master
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git switch -master
Switched to branch 'master'
```


QUESTION 3 — Feature Branch Development

```
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   health_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git add health_info.txt
warning: in the working copy of 'health_info.txt', LF will be replaced by CRLF the next time Git touches it
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git commit -m "Add project objective and course details"
[master 9adf74e] Add project objective and course details
1 file changed, 4 insertions(+)
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git log --oneline
9adf74e (HEAD -> master) Add project objective and course details
a124286 Initial commit for health appointment system
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git status
On branch master
nothing to commit, working tree clean

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch appointment-feature
Switched to branch 'appointment-feature'
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch --show-current
appointment-feature
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ touch src/appointment.txt
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ nano src/appointment.txt
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git add src/appointment.txt
warning: in the working copy of 'src/appointment.txt', LF will be replaced by CRLF the next time Git touches it
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git commit -m "Add appointment module"
[appointment-feature 5d7e071] Add appointment module
1 file changed, 11 insertions(+)
create mode 100644 src/appointment.txt
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch
  appointment-feature
* master
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git switch -master
Switched to branch 'master'
```

```
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch --show-current
master
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git log appointment-feature --oneline
5afd701 (appointment-feature) Add appointment module
9adf74e (HEAD -> master) Add project objective and course details
e12428d Initial commit for health appointment system
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
```

```
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git add src/appointment.txt
warning: in the working copy of 'src/appointment.txt', LF will be replaced by CRLF the next time Git touches it
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git commit -m "Add appointment module"
[appointment-feature 5d7e071] Add appointment module
1 file changed, 15 insertions(+)
create mode 100644 src/appointment.txt
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch
* appointment-feature
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git log appointment-feature
5afd701 (HEAD -> master, appointment-feature) Add project objective and course details
e12428d Initial commit for health appointment system
```

QUESTION 4 — Merging and Conflict Resolution

```
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (system-update)
$ git switch master
Switched to branch 'master'

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ nano health_info.txt

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git add health_info.txt

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git commit -m "Update project name in master"
[master cfb3e3] Update project name in master
1 file changed, 3 insertions(+)

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git merge system-update
Auto-merging health-info.txt
CONFLICT (content): Merge conflict in health_info.txt
Automatic merge failed; fix conflicts and then commit the result.

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   health_info.txt

no changes added to commit (use "git add" and/or "git commit -a")

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ nano health_info.txt

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ git add health_info.txt

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ git commit -m "Updated project name in system-update"
[system-update 4fbeb9c] Update project name in system-update
1 file changed, 4 insertions(+)

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ git merge --abort
```

```
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   health_info.txt

no changes added to commit (use "git add" and/or "git commit -a")

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ nano health_info.txt

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ grep -n "<====<|>====>|" health_info.txt
12:<<<<<< HEAD
14:=====
19:>>>>>> system-update
21:=====
23:>>>>>> system-update

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git add health_info.txt

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git commit -m "Resolve health info merge conflict"
[master b105d9c] Resolve health info merge conflict
```


QUESTION 5 — Final Demonstration

```
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ find .
./
./git
./git/COMMIT_EDITMSG
./git/config
./git/description
./git/HEAD
./git/hooks
./git/hooks/applypatch-msg.sample
./git/hooks/commit-msg.sample
./git/hooks/fsmonitor-watchman.sample
./git/hooks/post-update.sample
./git/hooks/pre-applypatch.sample
./git/hooks/pre-commit.sample
./git/hooks/pre-merge-commit.sample
./git/hooks/pre-push.sample
./git/hooks/pre-rebase.sample
./git/hooks/pre-receive.sample
./git/hooks/prepare-commit-msg.sample
./git/hooks/push-to-checkout.sample
./git/hooks/sendemail-validate.sample
./git/hooks/update.sample
./git/index
./git/info/exclude
./git/logs
./git/logs/HEAD
./git/logs/refs
./git/logs/refs/heads
./git/logs/refs/heads/appointment-feature
./git/logs/refs/heads/master
./git/logs/refs/heads/system-update
./git/objects
./git/objects/07
./git/objects/07/851b1f481725074ec4403bb2e428b581bd07eb
./git/objects/13
./git/objects/13/df8c5142772cc05f8e0f538be180bed7ee5a18
./git/objects/3c
./git/objects/3c/70add63ee8279c8cd722882b055d345513664e
./git/objects/4f
./git/objects/4f/beb9c90ae3ab1c14ab777ef819a039ab63e71c
./git/objects/55
./git/objects/55/6dc77603483d1e9eb66ea37927230d0c32a732
./git/objects/5d
./git/objects/5d/7e071824eedb9d48cdfdc388815304350ba5fe
./git/objects/67
./git/objects/67/97b317ce04c548cade407546155c42c4ba7fa0
./git/objects/72
./git/objects/72/339288bbe31994d229229cd8f67554f5214ab6f
./git/objects/72/399f16508207766be1bec3dd34d4bf8a99c58
./git/objects/9a
```

```
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
./git/objects/f1
./git/objects/f1/a8a45e7ca10e02e36d11972ee374fbff463ea2
./git/objects/f9
./git/objects/f9/2cd3fb1d08620ad03c5b731930f1e1364511d6
./git/objects/fc
./git/objects/fc/6381096af595064dfa1d6cdb6fca2d97322286
./git/info
./git/pack
./git/refs
./git/refs/heads
./git/refs/heads/appointment-feature
./git/refs/heads/master
./git/refs/heads/system-update
./git/refs/tags
./backup
./config
./docs
./health_info.txt
./src
./src/appointment.txt

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git status
On branch master
nothing to commit, working tree clean

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git log --oneline --graph --all
* b105d9c (HEAD -> master) Resolve health info merge conflict
| * 4fbeb9c (system-update) Update project name in system-update
| * cfbdd31 Update project name in master
|/
* 5d7e071 (appointment-feature) Add appointment module
* 9adf74e Add project objective and course details
* e12428d Initial commit for health appointment system

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch
  appointment-feature
* master
  system-update

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ ls src
  appointment.txt

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ cat health_info.txt
Student Name: Ayush Bhargav
Roll Number: 241501877
```

```
DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch
  appointment-feature
* master
  system-update

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ ls src
  appointment.txt

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ cat health_info.txt
Student Name: Ayush Bhargav
Roll Number: 241501877

DIVYANSH NEGI@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ |
```